Experiential Learning Project Assessment-I on

"TOWER OF HANOI VISUAL SIMULATOR"

under the course

Python ProgrammingLanguage(N-PCCCM304P)

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Session:2026-27 (ODD)

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Problem statement

- Tower of Hanoi is a classic puzzle based on moving disks between three rods.
- The user selects the number of disks.
- All disks must be moved from the Source Rod to the Destination Rod.
- A larger disk cannot be placed on a smaller disk.
- The simulator displays each move step-by-step.
- It also calculates the minimum number of moves required.

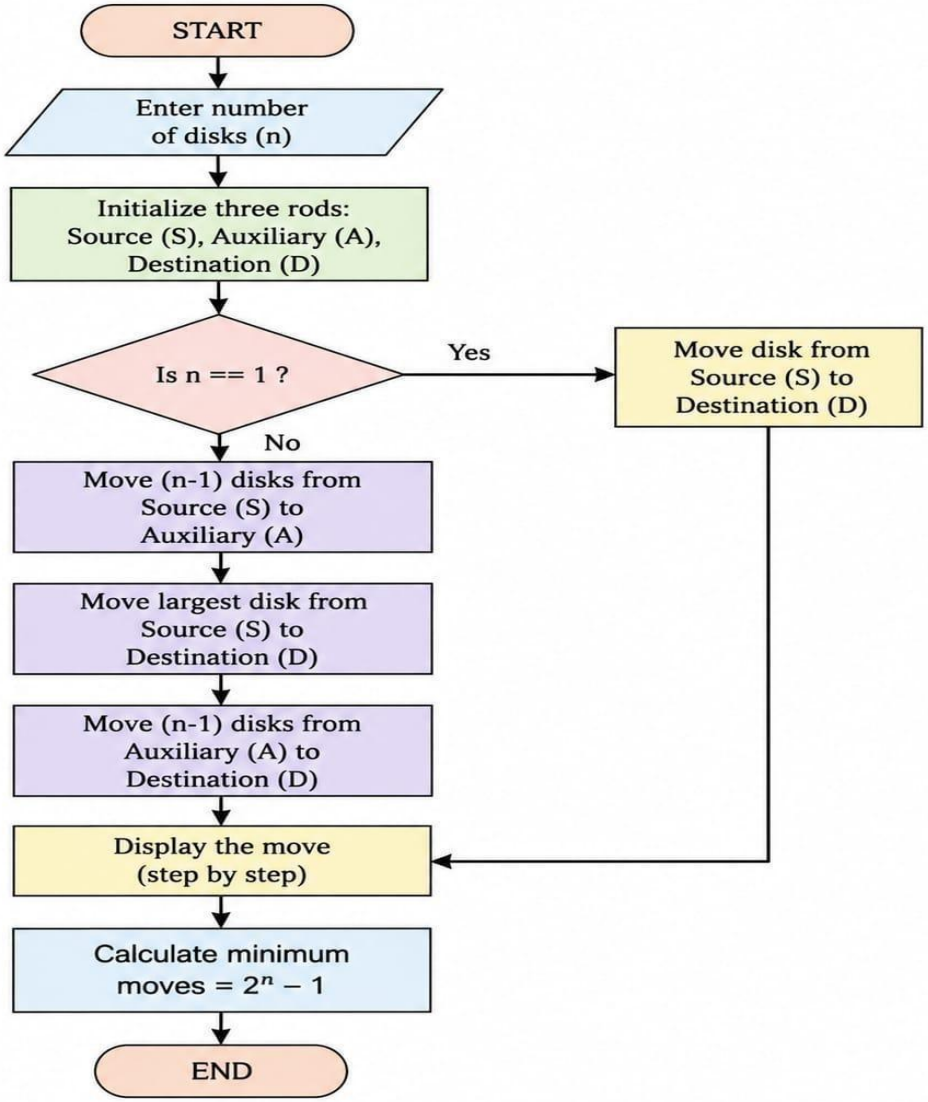
Objectives

- To understand the Tower of Hanoi problem.
- To implement the solution using Python.
- To use recursion for solving the puzzle.
- To visually display every disk movement.
- To calculate the minimum number of moves.
- To make the concept easy and interactive for users.

Algorithm

1. Start with n disks.
2. If there is only one disk, move it directly to the destination.
3. Move n-1 disks from Source to Auxiliary.
4. Move the largest disk from Source to Destination.
5. Move n-1 disks from Auxiliary to Destination.
6. Repeat until all disks are transferred.
7. Display every move sequentially.

Flowchart


START
Enter number of disks (n)
Initialize three rods: Source (S), Auxiliary (A), Destination (D)
Is n == 1 ?
Yes
Move disk from Source (S) to Destination (D)
No
Move (n-1) disks from Source (S) to Auxiliary (A)
Move largest disk from Source (S) to Destination (D)
Move (n-1) disks from Auxiliary (A) to Destination (D)
Display the move (step by step)
Calculate minimum moves = $2^n - 1$
END

Understanding of Python Logic

- Recursion: Solves the Tower of Hanoi problem.
- Functions: Separate the puzzle-solving and display operations.
- Loops: Used for displaying or processing moves.
- Lists: Can represent disks and rods.
- Conditional statements: Check the base case.
- GUI/Visualization: Can be used to show disk movement graphically.

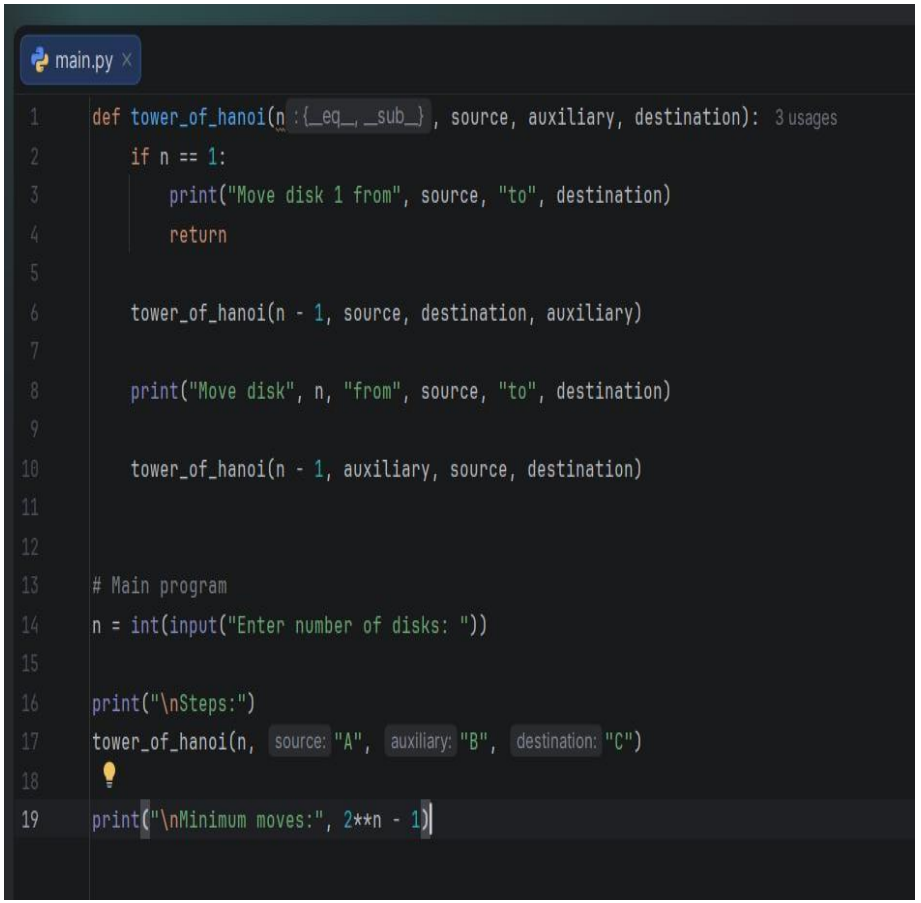
PHASE 2 : CODING & IMPLEMENTATION

Technology Used

Python

Implementation

- Take number of disks from the user.
- Use recursion to solve Tower of Hanoi.
- Store/display each move.
- Show the rods and disks visually.
- Calculate minimum moves using □



PROGRAM OUTPUT

- Enter the number of disks.
- The program starts the Tower of Hanoi process.
- Disks are moved from the Source rod to the Destination rod.
- A larger disk cannot be placed on a smaller disk.
- Each movement is displayed step-by-step.
- The minimum number of moves is calculated automatically.

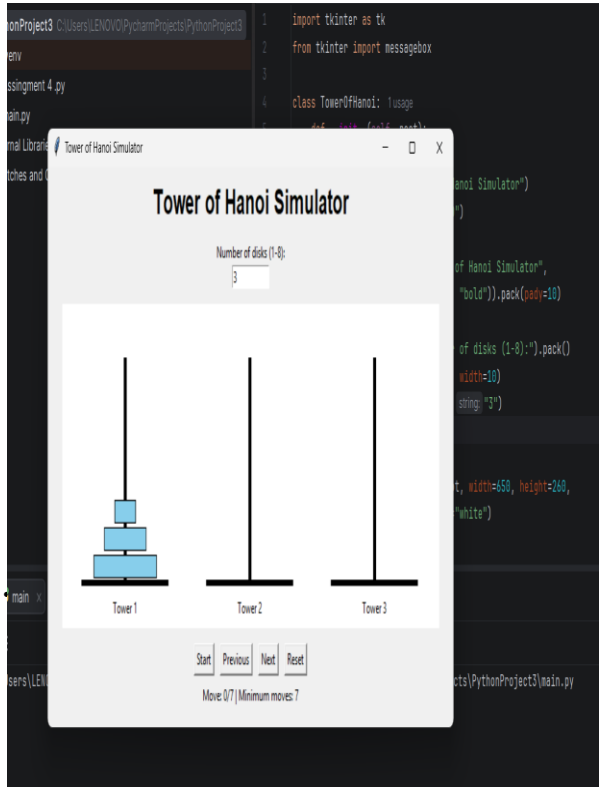


PHASE 3 : IMPLEMENTATION AND TESTING

- Implemented the Tower of Hanoi algorithm using Python.
- Used recursion to move disks between the three rods.
- Tested the program with different numbers of disks.
- Verified that a larger disk is never placed on a smaller disk.
- Checked whether all disks move correctly from the source rod to the destination rod.
- Verified the minimum number of moves using the formula $2^n - 1$.
- Tested the program to ensure correct output and error-free execution.

DEPLOYMENT

- Prepared the Python source code for execution.
- Uploaded the project code to a GitHub repository.
- Organized the project files for easy access and maintenance.
- Verified the program output after implementation.
- Made the project code accessible for learning and demonstration.
- Maintained the project repository for future updates.



GitHub Account, Repository Creation & Uploading Code

- Created a GitHub repository for the project.
- Uploaded the Python source code.
- Used meaningful commit messages.
- Updated the project regularly.
- Tested and corrected the program.
- Maintained individual contribution in the project.
- Github repository link

https://github.com/Tanvvviiii/Tower-of-Hanoi-Visual-Simulator/blob/main/tower_of_hanoi_visual_simulator_py

CONCLUSION

Successfully developed the Tower of Hanoi Visual Simulator using Python.

Implemented recursion to solve the puzzle efficiently.

Demonstrated disk movements between three rods step by step.

Calculated the minimum number of moves automatically.

Improved understanding of recursion, functions, loops, and conditional statements.

Created an educational tool to make the Tower of Hanoi problem easier to understand.

FUTURE SCOPE

- Add an interactive graphical user interface (GUI).
- Include animation to show disk movements in real time.
- Add Start, Pause, Resume, and Reset buttons.
- Allow users to select different numbers of disks.
- Display the current move count and minimum moves required.
- Add sound effects and customizable themes.
- Develop a web-based version for online access.
- Include a step-by-step tutorial to help beginners understand recursion.

Thank You